

ORIGINAL PAPER

The clustering lottery: initialisation and input-order sensitivity of mixture-model clustering confounds benchmarks of spatial domain identification

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Abstract

Motivation: New methods for spatial domain identification are usually compared by small differences in adjusted Rand index (ARI), mostly on the twelve human DLPFC Visium sections. Most of them obtain domains by fitting a Gaussian mixture model (typically mclust, EEE) once to a learned embedding. How much of the reported differences is due to the encoder and how much to this final clustering step is rarely measured.

Results: We evaluated two graph deep-learning methods (HiSTaR, GraphST), BANKSY, SpaGCN and training-free graph-diffusion baselines on 20 annotated sections from three technologies (10x Visium, STARmap, MERFISH), with three seeds each. With its official code, HiSTaR reached a median ARI of 0.51 on DLPFC instead of the reported 0.65. On 60 fixed embeddings, refitting the same mixture model from different random starts changed ARI by 0.17 on average and by up to 0.54; selecting the best of 20 fits by ARI would inflate results by 0.08. R mclust is deterministic for a given input, but merely permuting the order of the spots changed its ARI by up to 0.29, and adding noise at 1% of the feature scale by up to 0.53. Selecting fits by likelihood did not resolve this: likelihood and ARI were negatively correlated on MERFISH. Under matched clustering, neither deep model outperformed a training-free linear graph-diffusion baseline on any of the three datasets, and method rankings changed with the clustering protocol. We give five reporting recommendations for benchmarks.

Availability and implementation: Code, per-run results and figures: [TODO: GitHub URL and Zenodo DOI]. All public data sources are listed in the Data availability section.

Key words: spatial transcriptomics, spatial domains, clustering, reproducibility, benchmarking

Introduction

Spatial transcriptomics measures gene expression at thousands of locations in a tissue section while retaining their coordinates. Identifying *spatial domains*, regions with coherent expression such as cortical layers, is one of the first analyses of such data, and many methods have been proposed for it, from Bayesian models [Zhao et al., 2021] and neighbourhood-augmented features [Singhal et al., 2024] to graph neural networks [Hu et al., 2021, Dong and Zhang, 2022, Long et al., 2023] and hierarchical graph variational autoencoders such as HiSTaR [Yu et al., 2025].

Most of these methods are evaluated on the twelve manually annotated human dorsolateral prefrontal cortex (DLPFC) sections [Maynard et al., 2021] and report median ARI improvements of a few hundredths over previous methods. Many share the same last step: the learned embedding is clustered once with mclust’s EEE Gaussian mixture [Scrucca et al., 2016]. Benchmarks have found that rankings depend strongly on the dataset [Yuan et al., 2024, VERIFY, 2025a],

that deep methods vary with the random seed [VERIFY, 2025a], that replacing each method’s clustering algorithm changes results [Hu et al., 2024], and that preprocessing and clustering choices often matter more than architectural novelty [VERIFY, 2026]. Simple training-free pipelines are also competitive [VERIFY, 2025b, Do et al., 2026].

We complement these studies with a direct measurement of the variance contributed by the clustering step itself. We ask: (1) do HiSTaR’s published DLPFC results reproduce with the official code; (2) on a *fixed* embedding, how much does ARI change with the initialisation of the mixture model, and with perturbations that leave the data essentially unchanged (spot order, negligible noise) under R mclust; (3) does selecting fits by likelihood help; and (4) do encoders still differ once every embedding is clustered the same way, across three technologies? The study design is summarised in Fig. 1.

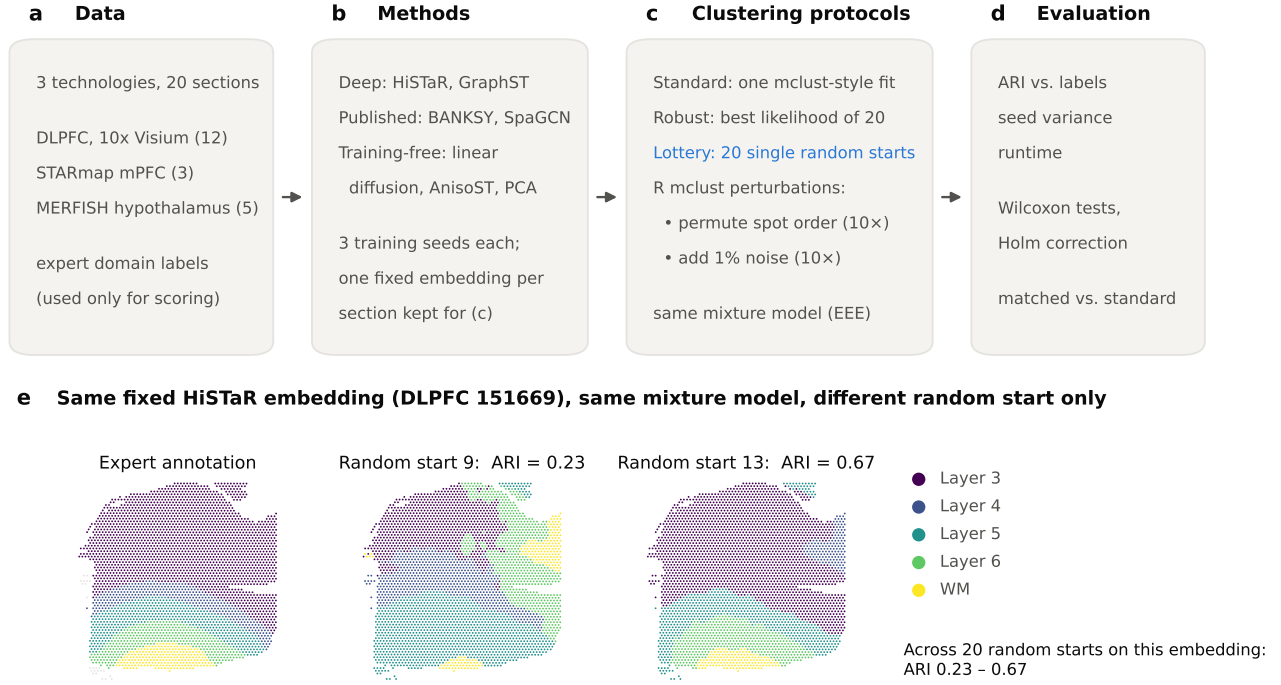


Fig. 1. Study design. **a–d:** data, methods, clustering protocols and evaluation. **e:** a real example of the clustering lottery. One fixed HiSTaR embedding of DLPFC section 151669 was clustered with the same tied-covariance Gaussian mixture from two different random starts; only the initialisation differs, yet ARI is 0.23 in one run and 0.67 in the other.

Materials and methods

Data

DLPFC (10x Visium). Twelve sections from three donors [Maynard et al., 2021]; seven annotated layers (five for donor 2), `layer_guess_reordered`. **STARmap mouse medial prefrontal cortex.** Three sections (about 1050 cells, 166 genes) with four annotated layers (L1, L2/3, L5, L6) [Wang et al., 2018]. **MERFISH mouse hypothalamus.** Animal 1, five sections from Bregma -0.04 to -0.24 (about 5500 cells, 155 genes) with eight annotated regions [Moffitt et al., 2018]. The STARmap and MERFISH annotations are those distributed with the BASS analyses [Li and Zhou, 2022]. In total, 20 sections from three technologies.

Preprocessing

DLPFC: genes in fewer than 50 spots or with fewer than 10 counts were removed, 2000 highly variable genes were selected (Seurat v3 flavour), counts were library-size normalised, log-transformed and scaled, and reduced to 200 principal components (PCs). STARmap and MERFISH (targeted panels): all genes, same normalisation, 50 PCs. All methods received the same input where they accept one; BANKSY used the log-normalised genes and SpaGCN and GraphST their own documented preprocessing. The true number of domains was given to every method, and unannotated spots were excluded from evaluation.

Methods

- **HiSTaR** [Yu et al., 2025]: PyTorch port of the official code with the published defaults (cross-level weight 0.3, 200+200 epochs).
- **GraphST** [Long et al., 2023]: official package and tutorial pipeline (600 epochs, 20-PC embedding, R mclust EEE; with and without its label refinement). Because of its CPU cost, GraphST was run with one seed on DLPFC and three seeds on STARmap and MERFISH.
- **BANKSY** [Singhal et al., 2024]: official `pybanksy`, 18 neighbours, $m_{\max} = 1$, $\lambda \in \{0.2, 0.8\}$, 20 PCs.
- **SpaGCN** [Hu et al., 2021]: official tutorial pipeline without histology ($p = 0.5$, Louvain resolution search to the true number of domains, 200 epochs).
- **Training-free baselines.** *Linear diffusion:* the top 30 PCs are smoothed on the spatial 6-nearest-neighbour graph, $H^{(t+1)} = (1 - \alpha)H^{(t)} + \alpha D^{-1} A H^{(t)}$, $\alpha = 0.5$, $T = 10$. *AnisoST:* the same with edge weights recomputed at every step, $w_{ij}^{(t)} = \exp(-\|h_i^{(t)} - h_j^{(t)}\|^2 / \kappa^{(t)})$ (Perona–Malik-type edge-preserving diffusion [Perona and Malik, 1990]; $\kappa^{(t)}$ the median squared edge difference); a similar idea has been used inside a deep model [Wang et al., 2024]. *PCA* without smoothing is the non-spatial reference. All baseline settings were fixed on DLPFC donor 1 before any other data were analysed, and none were tuned on STARmap or MERFISH.

Clustering protocols

All embeddings except SpaGCN’s (which clusters internally) were clustered with a tied-covariance Gaussian mixture, the mclust EEE model. *Standard:* each method’s usual call, i.e. R

mclust for GraphST and a 3-start scikit-learn GMM [Pedregosa et al., 2011] with a fixed seed for the others. *Robust*: 20 random starts, keeping the fit with the highest log-likelihood (label free). *Lottery*: one seed-0 embedding per section and method, clustered 20 times with a single random start each. *mclust perturbations*: R mclust (version 6.0.1, default hierarchical initialisation) on the same embedding (i) unchanged, (ii) after random permutation of the rows, 10 times, and (iii) after adding Gaussian noise with 1% of each feature’s standard deviation, 10 times.

Evaluation and statistics

ARI [Hubert and Arabie, 1985] against the manual annotation; NMI gave the same conclusions (code repository). For every (method, section) we average over seeds. Methods were compared with two-sided Wilcoxon signed-rank tests over sections, with Holm correction [Holm, 1979] within each family of comparisons. Runtimes are single-thread CPU wall-clock times.

Results

HiSTaR’s published DLPFC results did not reproduce

With the official code and defaults, HiSTaR reached a median ARI over the twelve DLPFC sections of 0.509 (mean 0.491, 95% bootstrap CI 0.454–0.525) with the 3-start GMM, and 0.449 (mean, seed 0) when its embedding was clustered with R mclust as in the official pipeline, compared with a reported median of 0.65 [TODO: verify the reported values (0.65 median; 0.718 on 151673) against the HiSTaR article]. Fixing a bug that applies random masking at inference did not change accuracy (0.487). Possible causes include per-section tuning of the loss weights (described in the HiSTaR supplement), unpublished preprocessing and the choice of seed; the following sections show that the clustering step alone produces differences of this size. [TODO: Report the outcome of contacting the authors.]

On a fixed embedding, the clustering step alone moves ARI by up to 0.54

Figure 2 shows the ARI of 20 single-start fits of the same mixture model on each fixed embedding. Across the 60 HiSTaR, GraphST and AnisoST embeddings, the ARI range within an embedding was 0.17 on average and reached 0.54 (HiSTaR, STARmap). The size of the effect depended on the embedding: average ranges were 0.21–0.23 for HiSTaR and AnisoST on DLPFC, 0.25–0.41 on STARmap and 0.11–0.15 on MERFISH, whereas GraphST’s 20-dimensional PCA embeddings were more stable (0.13 on DLPFC, 0.06 on STARmap, 0.03 on MERFISH). The best of the 20 fits by ARI exceeded the average single fit by 0.08: choosing a clustering run by its agreement with the labels, explicitly or by trying a few seeds, would inflate a reported result by about as much as the gap between most published methods.

R mclust is deterministic but not stable

mclust initialises EM by model-based hierarchical agglomeration, so one call on a fixed embedding always returns the same partition. That partition, however, depended on details that carry no information (Fig. 3). Randomly permuting the order of the spots changed ARI by 0.11–0.14 on average within an embedding on DLPFC (up to 0.29) and by 0.09–0.14 on

MERFISH; on DLPFC, 22–43% of the permutations moved ARI by more than 0.05 for HiSTaR and AnisoST, and 11% for GraphST. Adding noise at 1% of each feature’s standard deviation had similar effects, and on STARmap changed the ARI of HiSTaR embeddings by 0.39 on average (up to 0.53). On all three STARmap sections mclust’s default fit was degenerate for the AnisoST embedding, with one cluster holding about 95% of the cells and the others only 8–31 cells each (ARI 0.00–0.06), a failure the random-start fits did not show. Over all 60 embeddings, mclust’s default fit had a lower mean ARI (0.40) than the best-likelihood fit of 20 random starts (0.50) and reached a log-likelihood at least as high in only 42% of cases.

Choosing fits by likelihood is not a remedy

A natural fix is to run many starts and keep the most likely fit. It helped some embeddings and not others. Across all 20 sections, it raised HiSTaR’s mean ARI by 0.053 (14/20 sections, $p = 0.004$) and GraphST’s by 0.025 (13/20, $p = 0.11$), and did not change BANKSY $\lambda = 0.8$ (+0.003, 9/20, $p = 0.68$; negative on STARmap and MERFISH). The reason is visible in Fig. 2: likelihood and ARI were positively related on DLPFC and STARmap (median Spearman ρ 0.22–0.53) but negatively on MERFISH (−0.36 to −0.57), where the best-fitting mixture separates cell types rather than anatomical regions. Better optimisation of the mixture model is therefore not the same as better domains.

Under matched clustering, deep encoders did not outperform a training-free baseline

Table 1 and Fig. 4 compare all methods under both protocols. Linear graph diffusion had the highest or near-highest mean ARI on all three datasets (DLPFC 0.558, STARmap 0.705, MERFISH 0.497). Over the 20 sections it outperformed HiSTaR with standard clustering (+0.126, 17/20 sections, Holm-adjusted $p = 0.005$), GraphST with its own mclust clustering (+0.135, 14/20, adjusted $p = 0.016$) and with robust clustering (+0.110, 16/20, adjusted $p = 0.016$), and SpaGCN (+0.234, 20/20, $p < 0.001$). Even with its label refinement, GraphST did not reach linear diffusion on any dataset (DLPFC 0.548, STARmap 0.550, MERFISH 0.366). Differences to HiSTaR with robust clustering (+0.074, 14/20, adjusted $p = 0.069$) and to BANKSY $\lambda = 0.8$ (+0.048 to +0.051, adjusted $p = 0.069$) were not significant after correction. The ranking of the deep methods relative to the baselines changed with the clustering protocol (Fig. 4). The deep methods were also the least stable: HiSTaR’s seed-to-seed standard deviation of ARI was 0.047 on DLPFC and 0.18 on STARmap, against 0.003–0.04 for the training-free baselines, and they were the slowest (median 23 min per DLPFC section for GraphST and 1–2 min for HiSTaR on one CPU thread, against 3 s for linear diffusion).

Edge preservation helps at boundaries on Visium but not on single-cell data

On DLPFC, AnisoST matched linear diffusion in ARI (0.551 vs 0.558) and improved accuracy on spots at layer boundaries by 0.025 (9/12 sections, $p = 0.034$, uncorrected) without changing interior accuracy (Supplementary Fig. S2). On MERFISH it was clearly worse (0.363 vs 0.497): at single-cell resolution neighbouring cells often differ by cell type, so an edge-preserving rule blocks the smoothing that is needed to reveal

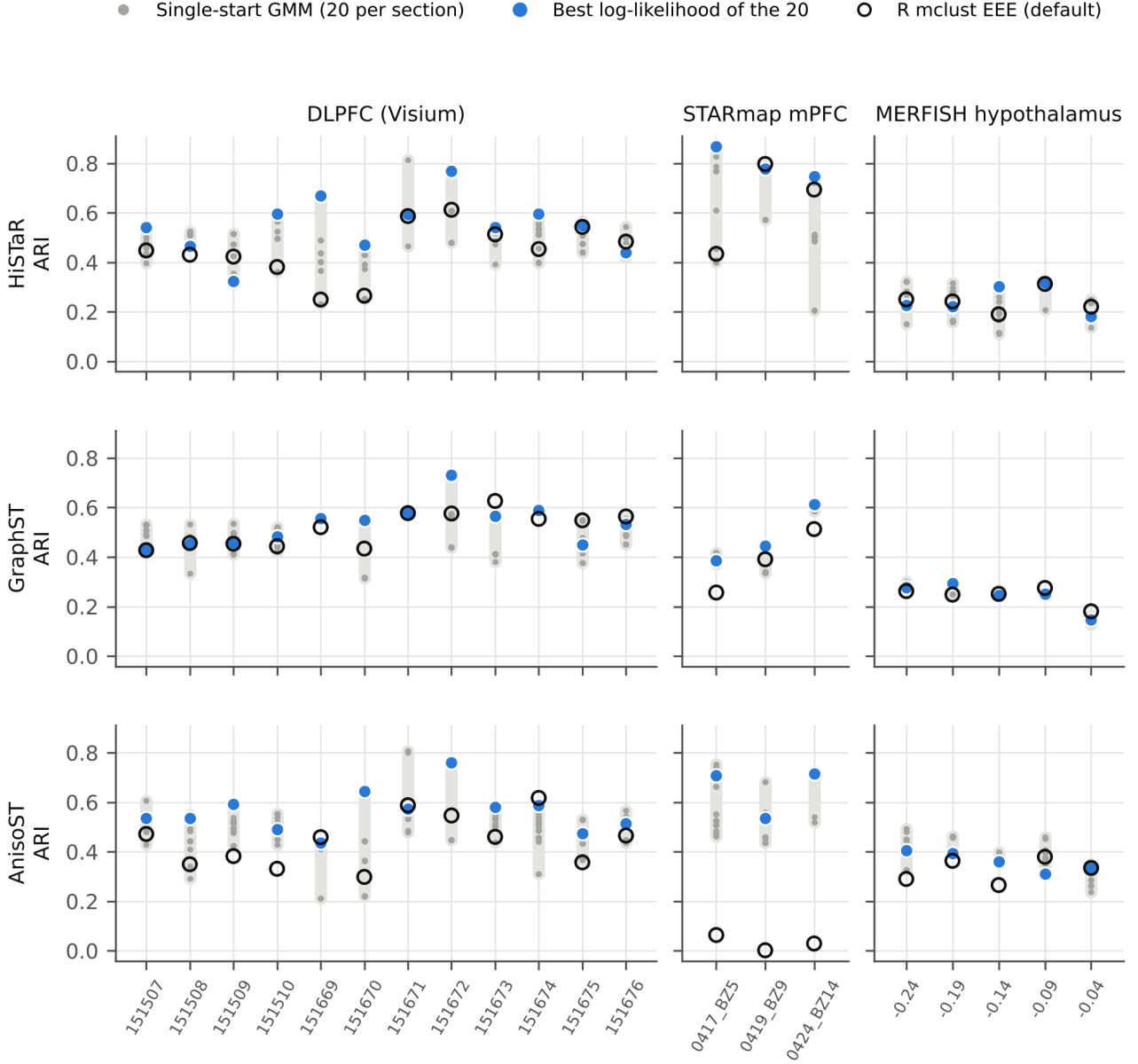


Fig. 2. The clustering lottery. For each section, one fixed embedding was clustered 20 times with a single-start tied-covariance GMM (grey dots; bar = range). Blue: the fit with the highest log-likelihood (label-free). Open circle: R mclust with default settings on the same embedding.

regions. We report this as a negative result for the edge-preserving variant; the simpler linear diffusion was the more robust baseline.

Discussion

The final clustering step of spatial domain methods is a substantial and under-reported source of variance. On fixed embeddings from three technologies, random initialisation alone spanned 0.17 ARI on average, and the R implementation used by most published methods changed its answer when the spots were merely reordered. Better likelihood did not reliably mean better domains, and under matched clustering a training-free diffusion baseline was not outperformed by

the deep encoders we tested. These results agree with recent large benchmarks [VERIFY, 2026, Hu et al., 2024, VERIFY, 2025a] and add a direct, quantitative measurement of clustering variance on fixed embeddings.

Recommendations.

(1) Report the clustering protocol: algorithm, implementation and version, initialisation, number of starts, seeds, and how they were chosen. (2) Never select seeds or runs by agreement with the labels. (3) Report distributions over training seeds *and* clustering runs (or input permutations), with per-section values. (4) Compare encoders under the same clustering, and include training-free baselines such as neighbourhood smoothing. (5) Report runtime and seed variance alongside accuracy.

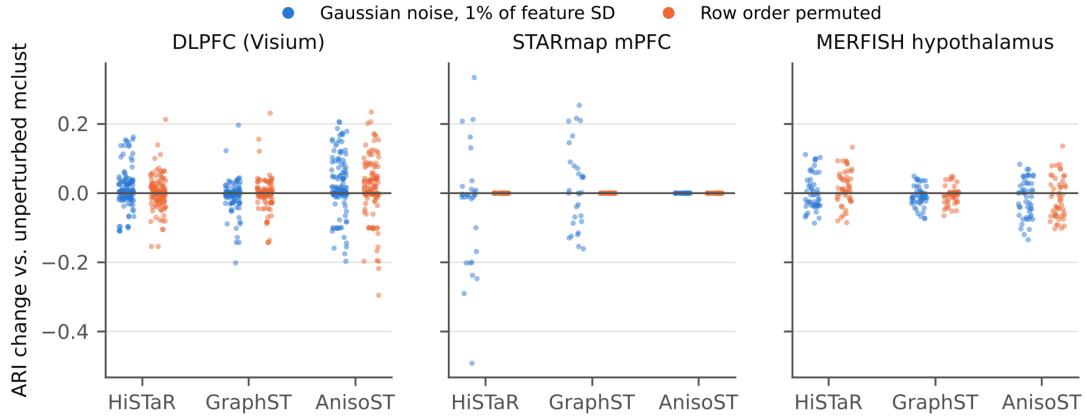


Fig. 3. Change in ARI of R mclust (EEE, default initialisation) when the same embedding is perturbed by permuting the spot order (orange) or adding noise at 1% of each feature's standard deviation (blue); 10 perturbations per section and embedding.

Table 1. Mean ARI over sections (3 seeds per section; GraphST on DLPFC: 1 seed). Standard: each method's usual clustering (R mclust for GraphST; 3-start GMM otherwise; SpaGCN clusters internally). Robust: 20-start GMM, best likelihood. [†]Training-free; all baseline settings fixed on DLPFC donor 1.

	DLPFC (Visium)		STARmap mPFC		MERFISH hypothalamus	
	Standard	Robust	Standard	Robust	Standard	Robust
Linear diffusion [†]	–	0.558	–	0.705	–	0.497
AnisoST [†]	0.526	0.551	0.564	0.622	0.377	0.363
PCA, non-spatial [†]	–	0.409	–	0.294	–	0.130
BANKSY $\lambda = 0.8$	0.494	0.518	0.740	0.694	0.428	0.410
BANKSY $\lambda = 0.2$	0.514	0.518	0.502	0.529	0.211	0.222
HiSTaR	0.491	0.544	0.593	0.658	0.221	0.264
GraphST	0.514	0.531	0.388	0.484	0.251	0.254
SpaGCN	0.422	–	0.267	–	0.151	–

Limitations.

We tested two deep encoders, BANKSY and SpaGCN; STAGATE, SEDR, BayesSpace and other methods were not included. GraphST was run with one seed on DLPFC. The STARmap and MERFISH annotations are coarse and the STARmap data contain only three small sections. The number of domains was given to every method. We could not reproduce HiSTaR's per-section tuning. The lottery experiments use one embedding per section and method; training-seed variance is reported separately.

Competing interests

No competing interest is declared.

Author contributions statement

[TODO: CRediT statement.]

Use of AI tools

[TODO: Describe per journal policy: code, experiments and a first draft of this manuscript were produced with the assistance of an AI coding assistant (Claude, Anthropic); all results were checked by the authors, who take full responsibility for the content.]

Data availability

DLPFC: spatialLIBD (count matrices) and SEDR analyses repository (layer annotations). STARmap and MERFISH with domain annotations: BASS-Analysis repository [Li and Zhou, 2022], from the original studies [Wang et al., 2018, Moffitt et al., 2018]. Scripts that download and convert all data are provided with the code.

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[TODO: Funding.]

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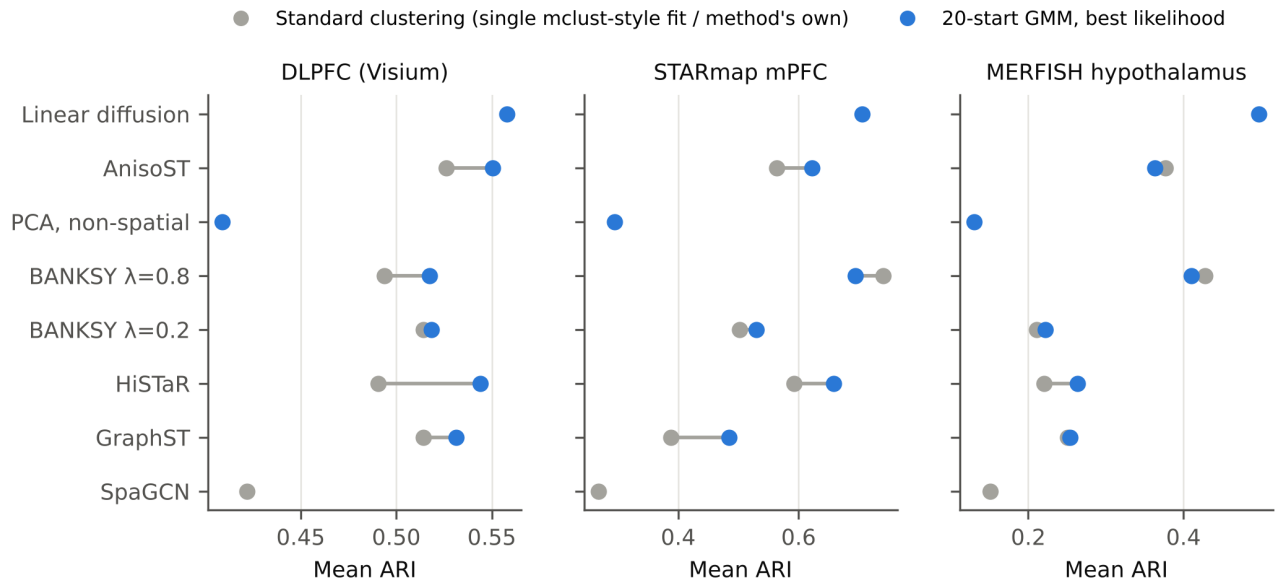


Fig. 4. Mean ARI per dataset with standard clustering (grey) and the 20-start, best-likelihood GMM (blue) applied to the same embeddings.

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Supplementary figures

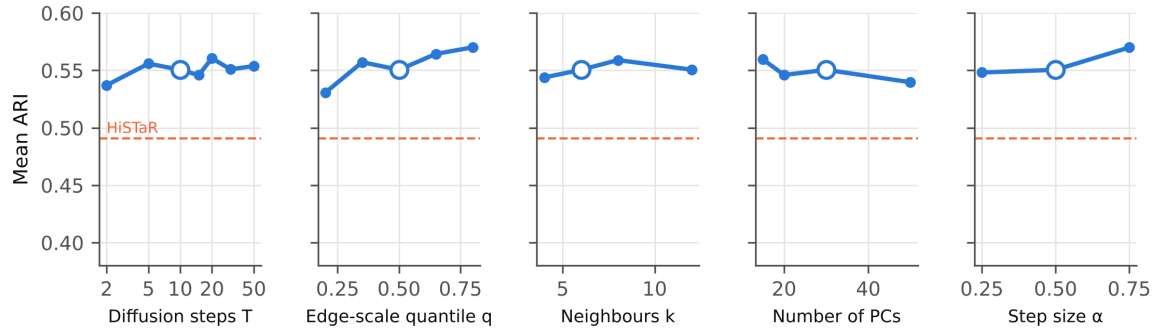


Fig. S1. DLPFC: mean ARI of AnisoST when one hyper-parameter is changed at a time; open circle = default; dashed = HiSTaR, standard clustering.

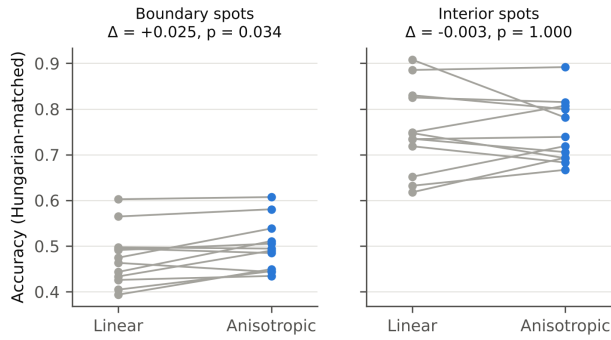


Fig. S2. DLPFC: accuracy on boundary and interior spots, linear vs. edge-preserving diffusion; one line per section.

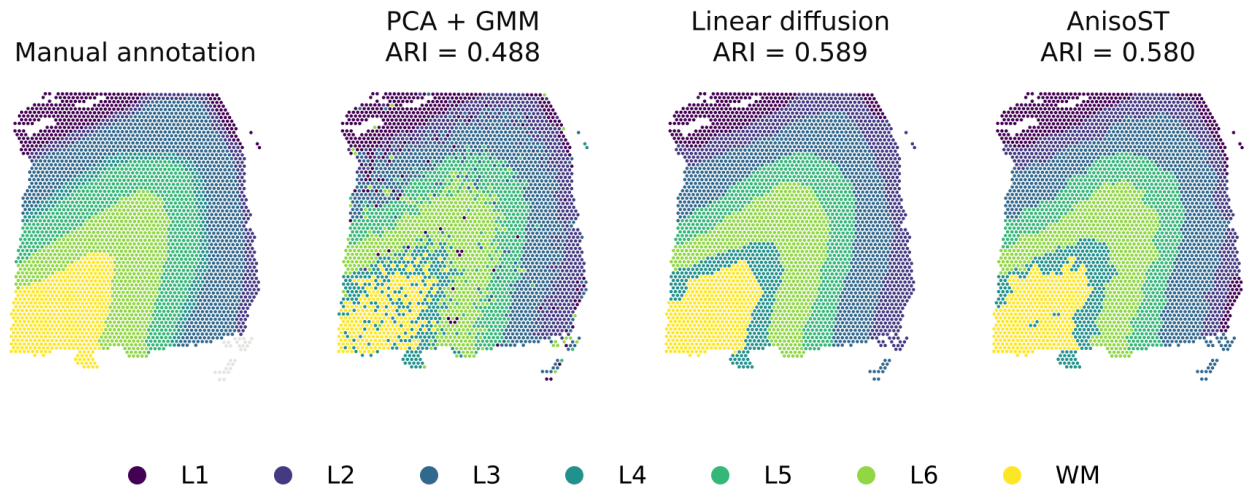


Fig. S3. DLPFC section 151673: manual annotation and domains from PCA + GMM, linear diffusion and AnisoST.